Derivations for the Prime Pages:
every formula used by the explainer and the explorables — derived in full, defined with source, or named as a gap

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2026-06-12

Abstract

The prime pages of this repository use a small set of formulas: wheel survival counts, the singular-series factors, the Hardy–Littlewood count, Sato–Tate moments and their gate variances, the Poisson and GUE baselines, the Brandt trace-extraction identities, the Eichler mass, the tapered prime spectrum, the Fibonacci-chain peak module, and the Chebyshev bias mechanism. This note derives each one at the level it deserves, under a strict three-bucket rule: *derived in full* (the derivation is complete on the page), *defined with source* (a deep theorem used as input, never re-proven and never assumed beyond its statement), or *named as a gap* (heuristic or open — said so). Nothing receives a fourth category. Eighteen numeric checks verify selected consequences, named per derivation in the text.

Scope. All mathematics here is classical; the derivations are expositions of known material, organised for the explorables, and we claim no priority for any of them. Heuristic steps are labelled at the point of use. Nothing in this note proves or claims progress on any open conjecture.

1 Part I: the wheels and the singular series

Derivation 1 (D1: wheel survival count)**.** *Fix a pattern of k distinct integer offsets $H = \{h_1, \dots, h_k\}$ and a prime p. A residue $r \bmod p$* survives *the wheel of p if $r + h_i \not\equiv 0 \pmod p$ for every i. The forbidden residues are exactly $\{-h_i \bmod p\}$, of which there are $w(p) := \#\{h_i \bmod p\}$ distinct values (offsets congruent mod p forbid the same cell). Hence exactly $p - w(p)$ residues survive, and the survival fraction is $1 - w(p)/p$. If $w(p) = p$ the pattern is* inadmissible*: every cell dies, and only finitely many occurrences can exist (each must include the prime p itself, as in $(n, n+2, n+4)$ at $p = 3$, realised once by $(3, 5, 7)$).*

Status: derived in full.

*Companion to the explorables (`explorables/`), the five-pictures explainer (`papers/how-the-primes-work.pdf`) and the ledger (`papers/prime-phenomena-ledger.pdf`) in `github.com/vfd-org/closure-positivity-lab`. Selected numeric consequences of the derivations (18 checks, named per derivation in the text) are verified by `python3 -m lab.derivations_check`; the machine-readable results ship at `out/derivations_check.json`.

Derivation 2 (D2: the independence baseline)**.** *The comparison model treats the k shifted values as unrelated integers. An integer avoids* $0 \bmod p$ *in $p-1$ of p residues, so k independent integers all avoid it with probability $((p-1)/p)^k$. This is a* model, *not a theorem about primes; it is the denominator against which the wheels' joint behaviour is measured.*

Status: derived in full (as a definition of the baseline).

Derivation 3 (D3: the per-wheel factor)**.** *The wheel of p corrects the independence baseline by*

$$\delta_p(H) \;=\; \frac{1 - w(p)/p}{(1-1/p)^k},$$

the ratio of the actual joint survival fraction (D1) to the independent one (D2). For all p exceeding every pairwise difference $|h_i - h_j|$ the offsets are distinct mod p, so $w(p) = k$.

Status: derived in full.

Derivation 4 (D4: the gap-ratio law)**.** *Specialise to pairs, $H = \{0, g\}$ with g even, and an odd prime p. If $p \nmid g$ then $w(p) = 2$ and $\delta_p = (1-2/p)(1-1/p)^{-2} = p(p-2)/(p-1)^2$ — the twin factor. If $p \mid g$ then $0 \equiv g$, so $w(p) = 1$ and $\delta_p = (1-1/p)(1-1/p)^{-2} = p/(p-1)$. The enhancement of a gap g over the twin gap, per odd prime $p \mid g$, is therefore*

$$\frac{p/(p-1)}{p(p-2)/(p-1)^2} \;=\; \frac{p-1}{p-2}, \qquad \text{so} \qquad w(g) \;=\; \prod_{\substack{p \mid g \\ p \text{ odd}}} \frac{p-1}{p-2}.$$

Hence $w(6) = 2$ (from p=3), $w(10) = 4/3$ (from p=5), and $w(210) = 2 \cdot \frac{4}{3} \cdot \frac{6}{5} = \frac{16}{5}$ (from $3, 5, 7$). These are the gap-slider's predictions, and the sieve measures 1.9950, 1.3318, 3.1994 *at 5×10^7 (ledger row 2).*

Status: derived in full.

Verified numerically: `D4` *in* `out/derivations_check.json`*.*

Derivation 5 (D5: the two forms of the twin constant)**.** $\dfrac{p(p-2)}{(p-1)^2} = \dfrac{(p-1)^2 - 1}{(p-1)^2} = 1 - \dfrac{1}{(p-1)^2}$, *so*

$$C_2 \;=\; \prod_{p>2} \frac{p(p-2)}{(p-1)^2} \;=\; \prod_{p>2} \Big(1 - \frac{1}{(p-1)^2}\Big) \;=\; 0.6601618158\ldots$$

The first form is the wheel product (D4); the second is the standard Hardy–Littlewood form. They are the same number by one line of algebra.

Status: derived in full.

Verified numerically: `D5` *in* `out/derivations_check.json`*.*

Derivation 6 (D6: convergence of the singular series)**.** *For p beyond the offsets' span, $w(p) = k$ and*

$$\log \delta_p = \log\Big(1 - \frac{k}{p}\Big) - k \log\Big(1 - \frac{1}{p}\Big) = -\frac{k(k-1)}{2p^2} + \frac{k - k^3}{3p^3} + O(p^{-4}),$$

by expanding both logarithms (the $1/p$ terms cancel exactly — this cancellation is the reason the product converges). Since $\sum_p p^{-2} < \infty$, the product $\mathfrak{S}(H) = \prod_p \delta_p(H)$ converges absolutely for every admissible pattern. For twins ($k = 2$) the second-order coefficient is -1, matching $1 - 1/(p-1)^2 = 1 - p^{-2} + O(p^{-3})$.

Status: derived in full.

Verified numerically: `D6, both expansion coefficients` *in* `out/derivations_check.json`*.*

Derivation 7 (D7: the Hardy–Littlewood count, and the lower limit)**.** Heuristic step, labelled as such. *In the Cramér model an integer near* t *is prime with probability* $1/\log t$*;* k *independent such events give* $1/\log^k t$*; the wheels correct the independence assumption place-by-place, multiplying by* $\mathfrak{S}(H)$ *(D3, D6). Summing over starting points,*

$$\#\{n \le x : n + h_i \text{ all prime}\} \;\approx\; \mathfrak{S}(H) \int_2^x \frac{dt}{\log^k t}.$$

This is *the Hardy–Littlewood* k*-tuple conjecture: the wheels prove admissibility and the constant; the asymptotic itself is open for every* $k \ge 2$ *(gap 1 of the mathematics page). Two facts about the integral that the explorables rely on: (i) any fixed lower limit* $A \ge 2$ *changes the integral by the constant* $\int_2^A dt/\log^k t$*, so all choices are asymptotically equivalent; (ii) at finite* x *and large* k *that constant is* not *small relative to the total (at* $k = 6$*,* $x = 2 \times 10^7$ *the* $[2, 100]$ *portion is over half), which is why the wheel-lab integrates from* 100 *and says so on the page.*

Status: heuristic, named as such; the asymptotic is the open conjecture.

Derivation 8 (D8: the Goldbach factor and the leading 2)**.** *Representations* $N = a + (N - a)$ *with both parts prime. At an odd prime* $p \nmid N$*:* a *must avoid* $\{0, N \bmod p\}$*, two distinct cells, giving the twin factor* $p(p-2)/(p-1)^2$*. At an odd* $p \mid N$*: the two forbidden cells coincide, giving* $p/(p-1)$ *— the same per-prime enhancement* $(p-1)/(p-2)$ *as in D4. At* $p = 2$ *with* N *even:* a *must be odd, survival* $1/2$*, against baseline* $(1/2)^2 = 1/4$ *— a factor of exactly* 2*. Collecting:*

$$R(N) \;\approx\; 2\,C_2 \prod_{\substack{p \mid N \\ p \text{ odd}}} \frac{p-1}{p-2} \int_2^{N-2} \frac{dt}{\log t \, \log(N-t)},$$

with the same heuristic status as D7. The leading 2 *in* $2C_2$ *— in the twin count as well — is the* $p = 2$ *wheel: even patterns survive the parity wheel twice as often as independence predicts.*

Status: derived in full (the factor structure; the count itself has D7's heuristic status).
Verified numerically: `D8` *in* `out/derivations_check.json`*.*

2 Part II: the instrument's gates

Derivation 9 (D9: Sato–Tate moments and the gate variances)**.** *For the measure* $d\mu = \frac{2}{\pi} \sin^2\theta \, d\theta$ *on* $[0, \pi]$ *and* $x = \cos\theta$*: using* $\cos^{2m}\theta \sin^2\theta = \cos^{2m}\theta - \cos^{2m+2}\theta$ *and* $\int_0^\pi \cos^{2j}\theta \, d\theta = \pi\binom{2j}{j}4^{-j}$*,*

$$I_{2m} := \int x^{2m} \, d\mu = 2\left[\binom{2m}{m}4^{-m} - \binom{2m+2}{m+1}4^{-(m+1)}\right] = \frac{\binom{2m}{m}}{(m+1)\,4^m} = \frac{\mathrm{Cat}(m)}{4^m},$$

since $\binom{2m+2}{m+1}/\binom{2m}{m} = 2(2m+1)/(m+1)$ *makes the bracket collapse to* $\binom{2m}{m}4^{-m}/(2(m+1))$*. So* $I_2 = \frac{1}{4}$*,* $I_4 = \frac{1}{8}$*,* $I_6 = \frac{5}{64}$*,* $I_8 = \frac{14}{256}$*; odd moments vanish by the symmetry* $\theta \mapsto \pi - \theta$*. The ledger's* 3σ *gates use the null variances of the empirical moments over* n *angles:* $\mathrm{Var}(\overline{x^j}) = (I_{2j} - I_j^2)/n$*, i.e.*

$$\sigma(m_1) = \tfrac{1}{2\sqrt{n}}, \quad \sigma(m_2) = \tfrac{1}{4\sqrt{n}}, \quad \sigma(m_3) = \sqrt{\tfrac{5}{64n}}, \quad \sigma(m_4) = \sqrt{\tfrac{10}{256n}},$$

which are exactly the constants used by `row07_sato_tate.py`*.*

Status: derived in full.
Verified numerically: `D9 (six checks)` *in* `out/derivations_check.json`*.*

Derivation 10 (D10: the Poisson and GUE baselines)**.** *Poisson (uncorrelated) unit-mean spacings are exponential:* $P(s < 0.2) = 1 - e^{-0.2} = 0.18127\ldots$*, the "0.18" of ledger row 10. The GUE Wigner surmise is* $p(s) = \frac{32}{\pi^2}s^2e^{-4s^2/\pi}$*; with* $a = 4/\pi$ *and the Gaussian moments* $\int_0^\infty s^3e^{-as^2}ds = \frac{1}{2a^2}$*,* $\int_0^\infty s^4e^{-as^2}ds = \frac{3}{8}\sqrt{\pi}\,a^{-5/2}$*:*

$$\mathbb{E}[s] = \frac{32}{\pi^2}\cdot\frac{\pi^2}{32} = 1, \qquad \mathbb{E}[s^2] = \frac{32}{\pi^2}\cdot\frac{3\sqrt{\pi}}{8}\Big(\frac{\pi}{4}\Big)^{5/2} = \frac{3\pi}{8},$$

so the surmise variance is $3\pi/8 - 1 = 0.17810\ldots$ *— the "0.178" gate. The surmise itself is an approximation to the exact GUE spacing law (defined with source: Mehta); the ledger gates against the surmise with the documented finite-height allowance.*

Status: derived in full (from the stated surmise density).
Verified numerically: `D10 (three checks)` *in* `out/derivations_check.json`*.*

Derivation 11 (D11: Eisenstein eigenvalue and trace extraction)**.** *The Brandt matrix* $B(P)$ *at a good prime* P *has constant row sums* $\mathrm{N}(P) + 1 = \#\mathbb{P}^1(\mathcal{O}_K/P)$ *(each ideal class receives all* $\mathrm{N}(P) + 1$ *subideals of norm* $\mathrm{N}(P)$*, weighted), so the all-ones vector is an eigenvector with eigenvalue* $\mathrm{N}(P)+1$*: the Eisenstein class. With Brandt dimension* $h = 2$ *the remaining eigenvalue is read off the trace,* $a_P = \mathrm{tr}\,B(P) - (\mathrm{N}(P) + 1)$*. With* $h = 3$ *and a genus-2 cuspidal block whose eigenvalues are a Galois pair* $\alpha, \sigma(\alpha) \in \mathbb{Z}[\varphi]$*:*

$$\mathrm{Tr}(\alpha) = \mathrm{tr}\,B - (\mathrm{N}(P) + 1), \qquad \mathrm{N}(\alpha) = \frac{\det B}{\mathrm{N}(P) + 1},$$

since trace and determinant collect eigenvalues additively and multiplicatively. The test for real multiplication: α *generates* $\mathbb{Q}(\sqrt{\Delta})$ *with* $\Delta = \mathrm{Tr}^2 - 4\mathrm{N}$*, so* $\alpha \in \mathbb{Z}[\varphi] \setminus \mathbb{Z}$ *iff* $\Delta = 5 \times (\text{square}) > 0$*. These are the identities used throughout* `realization/route_b/`*.*

Status: derived in full (given the standard Brandt row-sum property, defined with source: Eichler/Pizer).
Verified numerically: `D11` *in* `out/derivations_check.json`*.*

Derivation 12 (D12: the Eichler mass by orbit–stabiliser)**.** A_5 *(order 60) acts on* $\mathbb{P}^1(\mathbb{F}_q)$ *(*$q+1$ *points); the ideal classes are the orbits. The mass is*

$$\sum_{\text{orbits}} \frac{1}{|\mathrm{Stab}|} = \sum_{\text{orbits}} \frac{|\text{orbit}|}{60} = \frac{q+1}{60},$$

by orbit–stabiliser and the fact that the orbits partition the $q + 1$ *points. At* $q = 31$*: orbit sizes* $(12, 20)$*, stabilisers* $(5, 3)$*, mass* $\frac{1}{5} + \frac{1}{3} = \frac{8}{15} = \frac{32}{60}$*. The identity* $\sum 1/|\mathrm{Stab}| = (q + 1)/60$ *holds for any* q *by the same two lines; the orbit sizes quoted are the computed ones at* $q = 31$*. The mass check in the realization is this identity — a structural sanity gate, automatic once the orbit count is right.*

Status: derived in full.
Verified numerically: `D12` *in* `out/derivations_check.json`*.*

3 Part III: the spectra

Derivation 13 (D13: why the tapered prime sum peaks at the zeros)**.** *The crystal page plots* $F_X(k) = \big|\sum_{p\le X} \frac{\log p}{\sqrt{p}}\big(1 - \frac{\log p}{\log X}\big)p^{ik}\big|$*. The mechanism:* F_X *is (up to prime-power corrections)* *a Riesz-smoothed partial sum of the Dirichlet series of $-\zeta'/\zeta$ at $s = \frac{1}{2} - ik$, whose analytic continuation has a simple pole at every non-trivial zero; at $s = \frac{1}{2} - ik$ with $k > 0$ the nearby pole is the conjugate zero $\frac{1}{2} - i\gamma$, so by the conjugation symmetry of the zeros the peaks appear at $k = |\gamma|$, i.e. at the ordinates. Smoothing converts each nearby pole into a finite peak of width $\sim 1/\log X$, and the taper suppresses the Gibbs ringing that a sharp cutoff produces. We do not derive the full asymptotic here (gap: making the peak shape and height uniform in k and X is genuine analytic work we have not done); what we assert is the checkable consequence, and we check it: every zero in the plotted range owns a local peak, at two independent prime cutoffs ($X = 10^6$ in the shipped figure, 13/13; $X = 2 \times 10^5$ re-verified independently in the check suite, 13/13), with the sharp-cutoff control failing (5/13) exactly as the Gibbs explanation predicts.*

Status: named as a gap (the uniform peak asymptotic); the mechanism above is heuristic and its checkable consequence is verified.
Verified numerically: `D13` in `out/derivations_check.json`.

Derivation 14 (D14: the Fibonacci-chain peak module)**.** *The chain is built by the substitution $A \to AB$, $B \to A$ with tile lengths φ and 1. That its diffraction is pure point with Bragg peaks on a rank-2 module is the standard cut-and-project theory of the Fibonacci quasicrystal (defined with source: Baake–Grimm,* Aperiodic Order *I). The checkable consequence: the strongest peaks of the computed spectrum lie at $k = \frac{2\pi}{\bar{\ell}}(m + n\varphi)$ for small integers m, n, where $\bar{\ell}$ is the mean tile length. The check suite locates the eight strongest peaks of the shipped 3,000-tile spectrum and finds every one within 0.02 of that module.*

Status: defined with source (the consequence checked numerically).
Verified numerically: `D14` in `out/derivations_check.json`.

Derivation 15 (D15: the Chebyshev bias mechanism)**.** *Let χ be the non-trivial character mod 4 and $\psi(x,\chi) = \sum_{n \le x} \Lambda(n)\chi(n)$, $\theta(x,\chi) = \sum_{p \le x} (\log p)\chi(p)$. Splitting ψ over prime powers: the square terms contribute $\sum_{p^2 \le x} (\log p)\chi(p^2) = \sum_{p \le \sqrt{x}} \log p \cdot \chi(p)^2 = \theta(\sqrt{x}) \sim \sqrt{x}$, since $\chi(p)^2 = 1$ for every odd p —* all prime squares land in the residue class 1 mod 4*. Cubes and higher contribute $O(x^{1/3} \log x)$. Hence*

$$\theta(x,\chi) \;=\; \psi(x,\chi) \;-\; \sqrt{x}\,(1 + o(1)),$$

and since $L(s,\chi)$ has no pole, $\psi(x,\chi)$ is purely a sum of zero oscillations. Partial summation then gives

$$\pi(x;4,3) - \pi(x;4,1) \;=\; \underbrace{\frac{\sqrt{x}}{\log x}\,(1 + o(1))}_{\textit{the square-term contribution}} \;+\; (\textit{oscillations from the zeros of } L(s,\chi)),$$

where the labelled term is the contribution of the prime squares — not a dominance asymptotic, since the oscillation term is of the same order. This derives the mechanism *and the scale: the bias coefficient $D(x)\log x/\sqrt{x}$ is $O(1)$ (measured: 2.03, 1.11, 0.57 at 10^6, 10^7, 5×10^7 — order one and fluctuating, as the oscillation term predicts). What it does* not *derive is dominance: the oscillations are of the same order as the main term, so "the 3-class leads almost always" requires the logarithmic-density analysis of Rubinstein–Sarnak under GRH + linear independence — named as a gap, exactly as on the race page.*

Status: derived in full (the square-term mechanism; dominance is the named gap).
Verified numerically: `D15` in `out/derivations_check.json`.

4 The gap ledger, cross-referenced

For the explorables and explainer, the open items above are exactly three of the mathematics page's six named gaps: the heuristic status of the Hardy–Littlewood asymptotic (D7, D8 $\rightarrow$ gap 1), the uniform asymptotics of the tapered spectrum's peaks (D13 $\rightarrow$ part of gap 6), and bias dominance (D15 $\rightarrow$ part of gap 6). Everything else used by the prime pages is either derived in full above or defined with an explicit source (Ramanujan bound; Sato–Tate law; exact GUE; Fibonacci cut-and-project; Brandt row-sum property). The verification artifact `out/derivations_check.json` certifies all eighteen checkable statements.

References

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